PharmaScan: Calibrated Open Set Pill Verification with Visual Embeddings and Imprint Evidence

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Abstract

For patients managing several medicines, verification of loose pills involves recognising newly enrolled medicines while rejecting pills outside a personal medication list. PharmaScan formulates this task as retrieval from a personal gallery, with explicit outcomes for an accepted match, an unknown pill and a request for another photograph. A hybrid visual encoder is trained with an angular margin objective to produce a 256-dimensional embedding, through which each medicine is enrolled from three photographs without network retraining. Rejection and ambiguity thresholds are selected using separate validation medicines, while optical character recognition with Florence-2 supplies complementary imprint evidence. On 200 held-out appearance groups from the Computational Photography Project for Pill Identification (C3PI) collection, simulated galleries of ten medicines yield 76.0% correct naming and 0.04% wrong naming among enrolled queries, with 94.8% rejection of medicines outside the gallery. Imprint re-ranking increases identification accuracy among 4,936 catalogue products from 71.6% to 80.2%. Transfer to MEDISEG reveals a substantial acquisition gap, with 70.4% of enrolled queries rejected as unknown. These findings support calibrated verification under familiar imaging conditions and identify acquisition shift and imprint visibility as constraints on broader use. Our code is available at https://github.com/398-21/ultrasound_Model.

1 Introduction

Verification of loose tablets and capsules is a practical challenge in medication management because removal from packaging separates a medicine from its identifying information. Appearance then provides a recognition cue, although orientation, illumination and the visible side vary, and similar colours and shapes occur across products and strengths. Within the problem of medication errors addressed by the World Health Organization's Medication Without Harm initiative [27], visual verification concerns the association between a photographed pill and a known medicine before a medication decision is made.

A personal medication list creates an evolving recognition task in which newly prescribed medicines are enrolled from a few photographs and a query may belong to none of the enrolled identities. Recognition therefore depends on a representation that transfers across identities, together with a decision rule that distinguishes a plausible match from an unfamiliar or ambiguous pill. Since an incorrect accepted name and a request for further checking have different consequences, retrieval accuracy alone provides an incomplete account of verification performance. Existing pill retrieval benchmarks support representation learning [9, 22], while the personal workflow additionally involves membership rejection and handling of close candidates.

PharmaScan formulates this task as retrieval from a personal gallery with three outcomes: an accepted identity, UNKNOWN, or RETAKE. Medicines are registered through a shared visual embedding without network retraining, and decisions combine similarity to the nearest enrolment prototype with separation from the next candidate. Optical character recognition (OCR) using Florence-2 [28] supplies

complementary imprint evidence, while catalogue retrieval provides candidates when matching within the personal gallery is insufficient. Product name and strength remain associated with the pharmacy label [14], as visual similarity alone does not establish strength or suitability for administration.

The study investigates verification of previously unseen medicines from three enrolment photographs at an operating point selected on validation data, with particular attention to accepted misidentification, imprint evidence and acquisition transfer. Evaluation combines a strict identity-disjoint test from the National Library of Medicine (NLM) Computational Photography Project for Pill Identification (C3PI) collection [29], the ePillID benchmark [22], and MEDISEG images of Hong Kong products [2]. Their respective protocols, detailed in Section 4.1, distinguish transfer to new identities under familiar conditions from changes in cameras, product population and scene. The project plan's validation targets and corresponding measurements are reported separately from held-out results in Appendix C, Table A3.

The contribution is a calibrated verification workflow and an empirical analysis of the effects of its design choices. Sequential training changes, imprint re-ranking and rejected reader adaptations are examined through accepted errors and abstention, revealing acquisition shift as a constraint even when misidentification among enrolled queries is rare. The evidence is drawn from public image datasets and simulated galleries, with patient outcomes outside the evaluation scope.

2 Related Work

Pill identification and low-shot retrieval. Consumer-to-reference matching was established as a medication recognition task by the NLM Pill Image Recognition Challenge [29], followed by fine-grained pill identification using learned convolutional features and comparisons with handcrafted descriptors [25]. These studies demonstrate the usefulness of appearance while exposing the difficulty of separating similar products under image variation within a product. This setting is developed further by ePillID, whose strongest low-shot baseline combines bilinear features with multi-head metric learning [22]. Its analysis of confusing classes is relevant to personal galleries containing medicines with similar appearances, although ranked retrieval from product references leaves gallery membership rejection as a separate decision. PharmaScan evaluates both tasks and distinguishes results under strict and official protocols because access to references for test identities changes the information available during representation learning.

Multiple visual cues and feature fusion. Complementary visual cues are useful when discriminating pill characteristics are not equally visible across images. Ling et al. introduce CURE images with varied acquisition conditions, a segmentation network and a multi-stream recognition model trained using Batch All followed by Batch Hard mining [9]. Their combination of pill cues and difficult training examples relates to the broader problem of recognition beyond familiar photographs. In PharmaScan, convolutional neural network (CNN) features [21] are combined with vision transformer (ViT) features [4], bringing together local spatial structure and self-attention between image patches [23]. These complementary spatial assumptions also connect the architecture with the course treatment of vision transformers [13, p. 36]. Channel and spatial attention are provided by the Convolutional Block Attention Module (CBAM) [26], while capsule routing aggregates branch features through agreement [18]. The inherited architecture supplies the representation in Section 3.1, whose performance is evaluated within the complete verification workflow.

Metric learning for enrolment. Similarity representations separate enrolment from network training by mapping identity information to a space used for retrieval. FaceNet learns this geometry through triplet supervision [20], while CosFace and ArcFace impose margins in cosine and angular space, respectively [3, 24]. These objectives relate identity supervision to distances between examples, connecting with the course discussion of embedding geometry and matching views [12, pp. 38, 47–49]. PharmaScan uses ArcFace during training and prototypes during inference, allowing the deployed gallery to contain identities outside the training classifier. Balanced identity sampling [7] and evaluation protocol [15] are relevant to interpreting such embeddings because optimisation and sampling changes can affect apparent differences between objectives. Accordingly, the sequential analysis in Section 4.3 evaluates the combined training recipe without isolating an advantage attributable to the loss alone.

Open set decisions and confidence calibration. Open set recognition extends discrimination among known identities with a membership decision [19], for which score-based detection provides

a baseline [6]. Membership in PharmaScan is defined by the patient's enrolled gallery, so a medicine familiar to the encoder can still be foreign to the patient. The consequences of false acceptance motivate selection of the operating point on validation data and separate reporting of abstention, consistent with the course treatment of decision thresholds [11, p. 92]. Calibration concerns the relationship between reported probabilities and observed correctness: Guo et al. document poorly calibrated network confidence and examine post-processing corrections [5]. Here, logistic calibration [17] is fitted to catalogue retrieval evidence, while gallery rejection and ambiguity retain their own thresholds. These distinct purposes underlie the decision mechanisms in Sections 3.3 and 3.4.

Imprint evidence and acquisition transfer. Florence-2 represents multiple visual tasks through a common prompted model and supplies the pretrained imprint reader used here [28]. Hidden, small or debossed text limits the evidence available to this reader, while adaptation introduces a further distinction between text recovery and verification performance. Low-rank adaptation (LoRA) provides a parameter-efficient mechanism for adapting pretrained models [8], but an incorrect token can promote a visually similar product even when average recovery improves. The analysis of reader adaptation therefore measures accepted identities alongside recovered text. Acquisition transfer is examined with MEDISEG, which adds instance annotations for Hong Kong products photographed under conditions different from the US reference collection [1, 2]. Changes in scale, scene and product population extend the evaluation beyond retrieval within the source dataset to the transfer of both representation and calibrated decisions.

3 Approach

A detected pill is cropped, embedded by a hybrid CNN and ViT encoder, and compared by cosine similarity with the patient's enrolled medicines or with a product catalogue. Decision thresholds are calibrated on validation medicines only, and the OCR imprint reader adds complementary evidence.

The implementation combines the team's model code with components adapted from the ePillID benchmark code[1]. Development and report preparation are documented in Appendix E.

3.1 Network

The encoder in Figure 1 processes each 224×224 crop through a VGG16 branch [21] with CBAM channel and spatial attention [26] and a ViT-B/16 branch [4]. Local texture, colour and edge features from the CNN are combined with the ViT's global layout representation through capsule routing [18]. Each branch is projected to a 256-dimensional capsule, and initially equal coupling coefficients are updated towards the branch whose features agree with the fused vector. A small transformer encoder [23] then refines four 64-dimensional tokens derived from this vector, after which their mean is mapped by a linear head to the unit-length embedding $\mathbf{z} = f_\theta(x) \in \mathbb{R}^{256}$. Auxiliary heads predict shape and colour during training. The full layer configuration, pretrained checkpoints and initialisation are given in Appendix B, Table A2.

3.2 Training

Objective. The embedding is trained with ArcFace [3] over the 1,710 training appearance groups defined in Section 4.1, plus auxiliary losses:

$$\mathcal{L} = -\log \frac{e^{s\cos(\theta_y+m_a)}}{e^{s\cos(\theta_y+m_a)} + \sum_{j\neq y} e^{s\cos\theta_j}} + \lambda_s \mathcal{L}_{\text{shape}} + \lambda_c \mathcal{L}_{\text{colour}}, \tag{1}$$

with $s = 32$, $m_a = 0.3$ and $\lambda_s = \lambda_c = 0.2$, where θ_j is the angle between $\mathbf{z}$ and the j-th class weight, $\mathcal{L}_{\text{shape}}$ is a cross-entropy and $\mathcal{L}_{\text{colour}}$ a binary cross-entropy (unknown labels ignored). The margin warms up over the first three epochs $(0, 0.1, 0.2)$, then stays at 0.3.

Optimisation. Training follows the schedule with two phases in Table 1: the ViT and VGG blocks 1–3 are frozen during phase 1, allowing the new layers to adapt before all parameters are fine-tuned in phase 2. Each batch contains P groups with $K = 4$ photographs per group [7], and AdamW [10] uses separate learning rates for pretrained backbones, new layers and the randomly initialised ArcFace class table. This separation accommodates their different initialisations, while margin warm-up

[1] https://github.com/usuyama/ePillID-benchmark

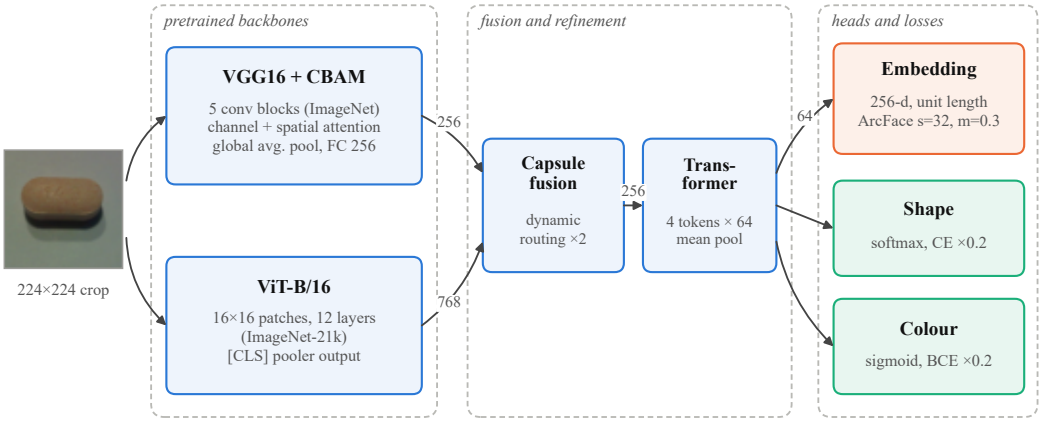


Figure 1: The hybrid network. VGG16 with CBAM attention and ViT-B/16 see the same crop. A capsule block fuses their features by dynamic routing and a transformer block refines the result into a unit-length 256-d embedding trained with ArcFace, with auxiliary shape and colour heads. Numbers on the arrows are feature sizes.

introduces angular separation progressively. The combined effect of these choices is evaluated in Section 4.3.

Table 1: Training settings of the served model (run 4). Selection: top-1 on medicines in the validation enrolment partition (best: phase 2, epoch 35). Seed 2026.

Setting	Phase 1	Phase 2
Trainable	all but ViT, VGG blocks 1–3	all
Max. epochs; batch ($P \times K$)	25; 32×4	35; 16×4
Learning rate: backbones / new / ArcFace	3×10^{-5} / 3×10^{-5} / 10^{-3}	10^{-5} / 10^{-5} / 10^{-4}
Optimiser	AdamW, weight decay 10^{-3}	same
Rate schedule	reduce on plateau ($\times0.5$, patience 3)	same
Early stopping	patience 6, best weights restored	same

Augmentation. Colour jitter is kept small (saturation 0.05, hue 0.0125) because colour contributes to identity. Crop windows are jittered by $\times0.75$–1.35 with centre shifts up to 10% to model detector error. Acquisition perturbations include brightness and contrast changes (±0.3), white balance ($\pm8\%$), downscaling to 40–160 pixels ($p = 0.3$), JPEG quality 30–90 ($p = 0.3$), and Gaussian noise ($\sigma = 0.03$, $p = 0.2$). A neighbouring pill is partly pasted into the frame with probability 0.2.

3.3 Open-set decision rule and calibration

A patient enrols each medicine j from k=3 photos. Its prototype is $\mathbf{p}_j = \sum_i f_\theta(x_{j,i}) / \|\sum_i f_\theta(x_{j,i})\|$. For a query with similarities $s_j = \mathbf{z}^\top \mathbf{p}_j$, sorted $s_{(1)} \geq s_{(2)}$,

$$\hat{y} = \begin{cases} \text{UNKNOWN} & s_{(1)} < \tau, \\ \text{RETAKE} & s_{(1)} \geq \tau \text{ and } s_{(1)} - s_{(2)} < m, \\ \arg\max_j s_j & \text{otherwise,} \end{cases} \tag{2}$$

where RETAKE requests another photograph with the pill turned over. Calibration uses validation medicines in 1,000 simulated ten-medicine galleries, yielding $\tau = 0.895$ for 95% rejection of foreign medicines and $m = 0.115$ as the largest margin with at most 10% retakes. The crop window is selected as $1.55\times$ the detected box's long side from a sweep of 1.3–2.0, across which correct naming varies by approximately one percentage point (Figure A4b).

3.4 Identification with imprints and calibrated confidence

For catalogue identification, a pill outside the personal gallery is ranked against 4,936 US products using each product's similarity to its closest reference as its appearance score s. Florence-2-large [28]

reads the imprint zero-shot at four rotations (Florence-2-base for inference on a central processing unit (CPU)). Each recorded imprint token receives an inverse document frequency (IDF) weight in $[0, 1]$, assigning greater weight to rare tokens such as 'EC5' than to common tokens such as '93', which appears on hundreds of Teva pills. A product's imprint score r is the weight of its best token found in the reads, and the top $K{=}100$ products by look are re-ranked by $s + \alpha r$ with $\alpha = 0.05$, both chosen on validation (Figure 4b). In the patient's gallery, only tokens that distinguish the enrolled imprints from each other confirm or contradict a match. Catalogue confidence is estimated by logistic calibration [17] from the top candidate's appearance score, its separation from the runner-up and its imprint evidence.

Raw cosine similarities are not probabilities: even products outside the catalogue can obtain similarities around 0.93. The fitted catalogue confidence is

$$\sigma(-12.50 + 12.50 s_1 + 72.13(s_1 - s_2) + 3.92 r_1)\,, \tag{3}$$

where s_1 and s_2 are the appearance scores of the first two re-ranked products and r_1 is the first product's imprint score. This estimate concerns correctness of catalogue candidates, while gallery membership and ambiguity remain governed by Equation 2.

3.5 Implementation and reproducibility

Training uses PyTorch [16] 2.11 and Hugging Face Transformers 5.19 on one NVIDIA RTX 4080 SUPER graphics processing unit (GPU) with 16 GB memory. The deployed model is served through FastAPI on a Hugging Face ZeroGPU Space[2]. Split generation, training, calibration and evaluation are separate scripts. The execution commands and output conventions are provided in Appendix E.

3.6 Software 3.0 in coding and evaluation

The evaluation harness was written with the coding agent from specifications in natural language, covering patient simulation, look-alike stress tests and comparisons of products from the same manufacturer, top-k and confidence evaluation, figure generation, and the API's 92 tests. Testing and review identified defects including an error in the label reader that linked an extended-release sticker to the immediate-release product, which was corrected by enforcing the release form. Procedural safeguards comprised selection of settings on validation data, one evaluation of each test set, and files generated by scripts as the source of reported numerical results.

4 Experiments

4.1 Data and evaluation setup

Training source and identity units. The primary source is the National Library of Medicine (NLM) Computational Photography Project for Pill Identification (C3PI) collection [29]. Its consumer-image list contains 40,882 entries, including 295 exact repeats, leaving 40,587 distinct photographs that were successfully cropped and assigned to partitions. Pill types are defined by nine-digit National Drug Code (NDC9) and product part, giving 2,400 types after package variants are consolidated. Types sharing recorded imprint, shape, colour and size form an *appearance group*, while those without imprint text remain separate. The resulting 2,338 groups define training labels and partition membership, with product identities retained for catalogue retrieval.

Consumer photographs vary in illumination, orientation, visible side and background, with a median of 15 images per medicine and a range of 3–101 in the inventory's NDC9 summary. Studio references provide more consistent views for catalogue matching and augment training groups in runs 3 and 4. Appendix A details the count reconciliation, package variants and grouping process, together with the distribution plot in Figure A3.

Partitioning and preprocessing. Groups containing an ePillID fold-4 holdout identity are reserved through NDC9 matching. Remaining groups are randomly divided approximately 80:10:10 between training and the two validation partitions (Table 2). Code for split generation checks shared identifiers, exact duplicates and near-duplicate crops. Only references for training groups are added in runs 3 and 4. Each detected pill is framed in a square window of 1.55 times the long side of its bounding

[2] https://davidpansx-pharmascan-api.hf.space/ui/

Table 2: C3PI partitions after removal of repeated entries and successful cropping of all 40,587 distinct consumer photographs. Studio references add 6,198 training crops in runs 3 and 4.

Partition	Groups	Crops	Role
Training	1,710	29,471	Representation learning
Validation enrolled	214	3,556	Selection and enrolment
Validation unknown	214	3,555	Rejection calibration
Held-out test	200	4,005	Evaluation on unseen groups

box and resized to 224×224 pixels. VGG16 inputs use ImageNet channel statistics, while ViT inputs are scaled to $[-1, 1]$, with constants given in Appendix B. Training augmentation is specified in Section 3.2. Figure 2 provides representative views.

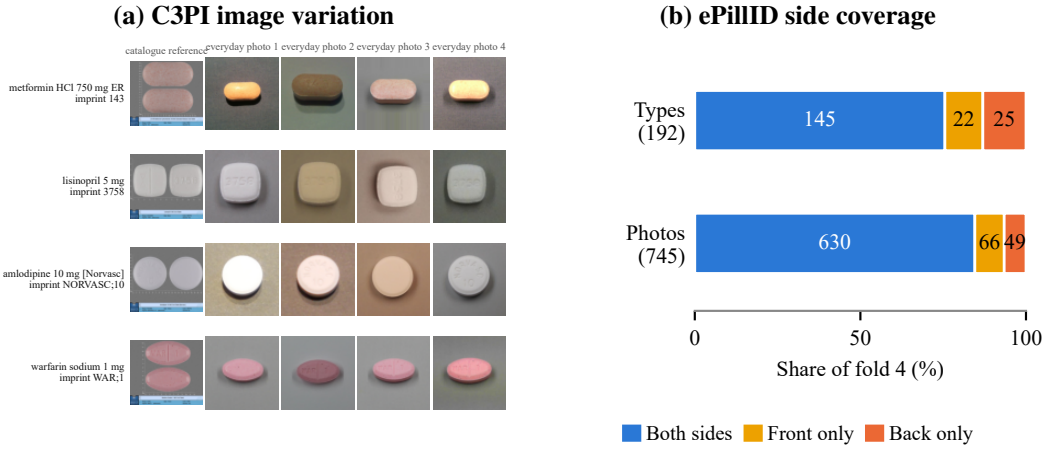


Figure 2: Data characteristics. (a) Catalogue references and four consumer photographs for each of four medicines illustrate variation in orientation, visible side and illumination. (b) ePillID fold-4 coverage by side availability, shown for pill types and photographs.

Evaluation sources and catalogue. The C3PI holdout comprises 200 appearance groups, whereas the local ePillID release contains 13,532 images of 4,902 types, including 9,804 references and 3,728 consumer images. Fold 4 contains 745 consumer photographs of 192 identities [22]. The official ePillID protocol permits reference images for test types during training, distinguishing it from the strict C3PI evaluation. MEDISEG [2] contributes 8,261 images with 14,069 annotated pills from 32 Hong Kong products. For evaluation with all products enrolled, three single-pill photographs per product form the gallery, leaving 13,973 query pills after 96 enrolment instances are excluded. The median pill long side is 81 pixels, with fifth and 95th percentiles of 27 and 194 pixels in 640×640 images. A separate deployment subset has 40 query pills. Product population, scale, cameras and clutter change together, so the external gap cannot be attributed to camera hardware alone. Catalogue identification uses 4,936 US products assembled from C3PI reference images and imprint, shape and colour metadata.

Data use. The study uses public image datasets without patient records. C3PI images are restricted to research and development, including when accessed through ePillID, whose MIT code licence does not supersede the image conditions. MEDISEG is distributed under CC BY 4.0, and download instructions for all sources are provided in `epills/docs/datasets.md`.

Metrics for medication verification. We simulate patients with 10 enrolled medicines (3 photos each). For the patient's own pills we report the share *named correctly*, *named wrongly* (an accepted misidentification), *retake* and *called unknown*; for pills that are not theirs, the share *rejected*. An accepted name can inform subsequent action, whereas unknown and retake outcomes add a further checking step to the workflow. We also report top-k accuracy, area under the receiver operating characteristic curve (AUROC), ePillID's mean average precision (MAP) and global average precision (GAP), and the reliability of the stated confidence. Intervals are obtained from 1,000 bootstrap

replicates, resampling medicine groups with replacement and retaining all their repeated query appearances. The 2.5th and 97.5th percentiles define the reported 95% intervals.

Gallery simulation. The evaluation code samples 1,000 galleries using seed 0. Within each gallery, ten eligible medicines are sampled without replacement and three enrolment images are selected per medicine. All remaining images of those medicines are used as enrolled queries. Enrolment images are excluded from queries within that gallery. Up to 200 images of medicines outside the gallery are sampled without replacement as foreign queries. Identities and photographs can recur across galleries. Percentages are computed by pooling query rows rather than averaging gallery percentages. For MEDISEG, enrolment is restricted to single-pill images. The simulation and bootstrap procedures are implemented in `epills/src/epills/training/eval/openset.py`; they were inspected for this report.

Baselines. (i) Run 1: the same network with a common recipe (new layers at 10^{-3}, no margin warm-up). (ii) The published ePillID models [22]. (iii) Look only versus look plus imprint, for the gallery and the catalogue.

4.2 Main results

Table 3 reports results under the strict protocol. In the conditions it was trained for, the served model (run 4) names 76.0% of a patient's own pills correctly and only 0.04% wrongly (95% confidence interval 0.02–0.07%), while rejecting 94.8% of pills that are not the patient's (Figure 3a). The imprint check raises correct names to 78.3% (wrong 0.06%) by settling retakes. These rates of wrong naming use enrolled queries as their denominator. With the imprint pipeline, 3.5% of foreign queries receive an accepted gallery identity and 94.8% are called unknown; the remainder request a retake (approximately 1.7%, inferred from rounded reported rates). Thus the 0.06% error among enrolled queries is not the overall rate of accepting a wrong medicine. Catalogue identification improves from 71.6% to 80.2% top-1 and from 91.7% to 94.4% top-5 with imprint re-ranking. Adding training-group references in run 3 raises strict ePillID MAP@1 from 13.7% to 37.1%. Run 4's acquisition augmentation gains 6.4 percentage points on MEDISEG while reducing C3PI top-1 by 1.2 points. Run 4 is the served model.

Table 3: Test results on medicines never seen in training (%; 10-medicine galleries, 3 enrolment photos, calibrated τ and m). Bottom block: the served pipeline with imprint reading (GPU reader, CPU reader in brackets).

	Run 2	Run 3	Run 4 (served)
C3PI held-out: top-1 among 200 medicines	82.0	**84.4**	83.2
gallery: named correctly / wrongly	78.6 / 0.1	**80.4** / 0.0	76.0 / 0.04
other medicines rejected	94.2	94.1	**94.8**
MEDISEG: top-1 among 32 products	38.9	41.8	**48.2**
gallery: named correctly / wrongly	17.6 / 1.1	18.7 / 1.0	18.7 / **0.7**
own called unknown / other rejected	51.5 / 78.7	64.2 / 89.3	70.4 / **93.1**
ePillID strict (no test type in training): MAP@1	13.7	37.1	**42.3**
C3PI gallery + imprint check: named correctly / wrongly			78.3 (77.3) / 0.06
retake / called unknown			7.2 / 14.5
foreign: unknown / accepted match			94.8 / 3.5
Catalogue (4,936 products) + imprint: top-1 / top-5			80.2 (79.2) / 94.4
look only: top-1 / top-5			71.6 / 91.7

4.3 Ablations and hyperparameters

Sequential design changes are summarised in Appendix C, Table A4. The combined optimisation changes produce the largest observed improvement. With the new layers at 10^{-3} the embedding collapses to about 3 effective dimensions and validation top-1 stalls at 77%, while the revised recipe keeps 12–15 dimensions and reaches 98% (Figure A5). Effective dimensionality is defined as $\exp(-\sum_i q_i \log q_i)$, where q_i is the fraction of embedding variance along principal direction i. It equals 256 for equal variance and one for a single direction. The decision settings are smooth (Figure A4): the crop window moves correct names by under one point, and the margin trades correct

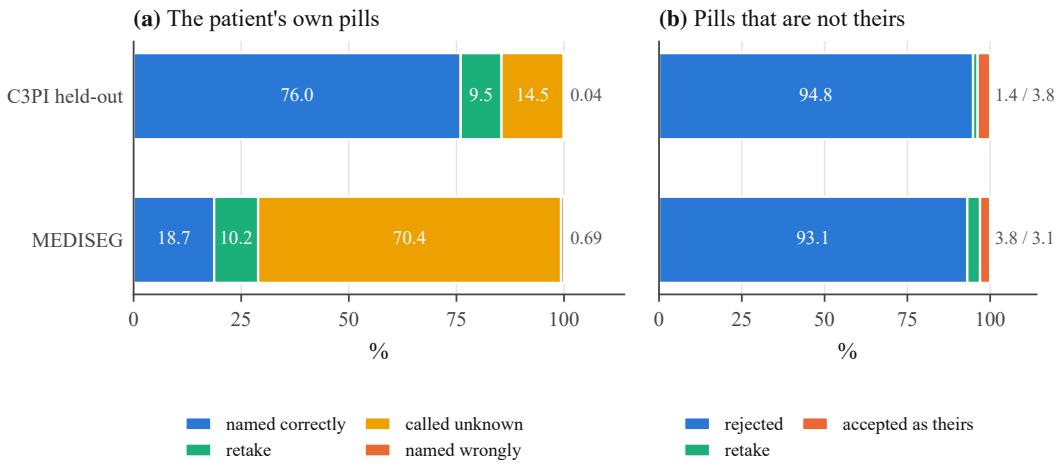


Figure 3: What the patient sees (run 4, embedding only, 10 medicines, 3 enrolment photos). Small shares are listed after each bar in legend order. In C3PI conditions most own pills are named correctly and wrong names are rare; under acquisition shift the threshold transfers poorly and most own pills are called unknown, which limits useful recognition.

names for retakes almost one for one. The imprint weight has a clear optimum (Figure 4b), because a large α lets false OCR reads promote wrong products. The imprint helps most when the pill's own imprint is actually read: the correct product ranks first in 92.9% of those queries, compared with 68.6% when it is not (Figure 4a).

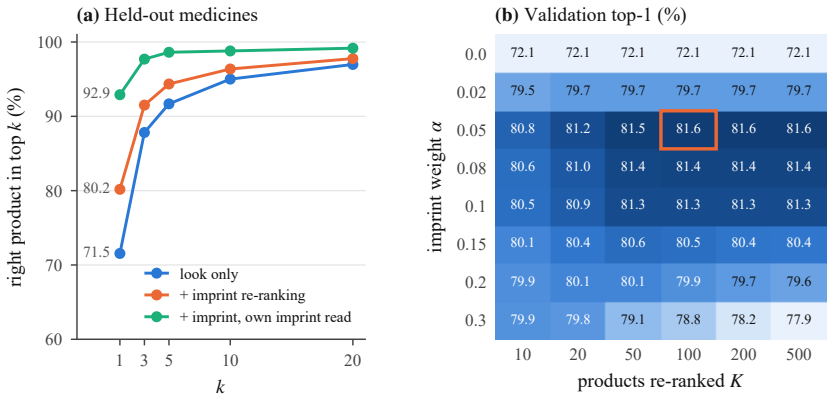


Figure 4: Catalogue identification. (a) Top-k on held-out medicines: the imprint helps most when the pill's own imprint is read (92.9% first vs. 68.6% when it is not). (b) Validation grid for α and K; the box marks the served setting.

4.4 Acquisition transfer and catalogue confidence

Acquisition shift produces the largest loss of useful recognition (Table 3, Figure 3). On MEDISEG, top-1 accuracy falls to 48% and the C3PI-calibrated threshold rejects 70% of enrolled queries, even though misidentification among enrolled queries remains below 1%. This decline is associated with image scale: top-1 accuracy is 36–37% for pills below 100 px and approximately 62% above that size (Figure 5a), while additional enrolment photographs provide only partial improvement (Figure 5b).

Catalogue verdicts also differ across sources, with 69.8% of MEDISEG pills receiving *no confident match* and 7.0% receiving *possible match*, compared with 75.8% *possible match* for catalogue pills. The relationship between stated confidence and observed correctness is shown in Figure 6. Deployment with the smaller CPU reader reduces identification accuracy by one percentage point to 79.2%, with no increase in the reported rate of wrong naming in the gallery (Table 3).

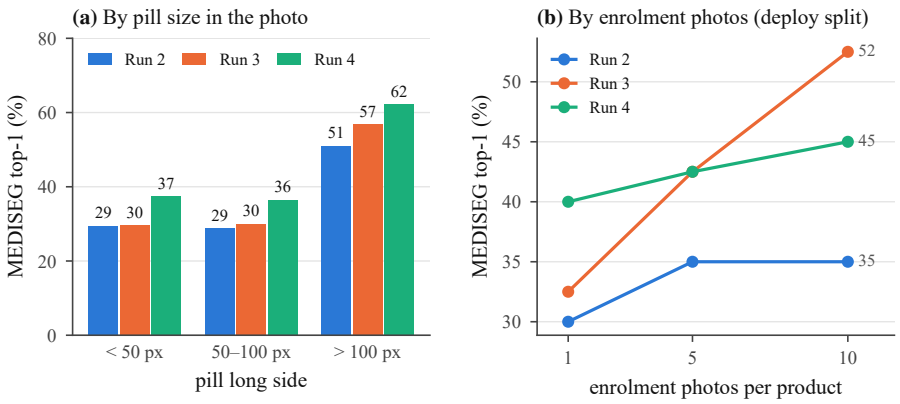


Figure 5: Robustness on MEDISEG (32 Hong Kong products, other phones, none in training). (a) Top-1 by pill size in the photo. (b) MEDISEG's deploy split (40 query pills): more enrolment photos help only partly.

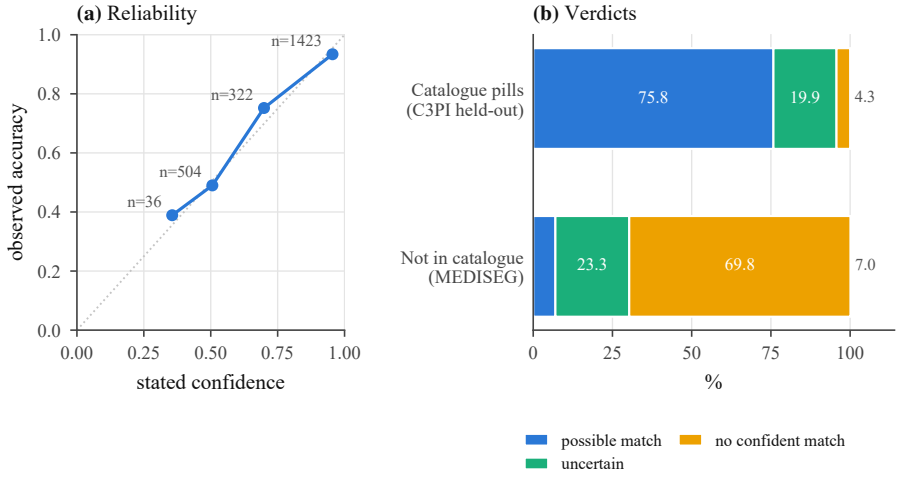


Figure 6: Calibrated confidence for identification. (a) Stated confidence against observed accuracy on held-out medicines. (b) Verdicts for catalogue pills and for pills not in the catalogue (MEDISEG).

4.5 Comparison with the ePillID benchmark

The two-sided evaluation averages scores across each front–back pair and includes 626 pairs from 145 types with both sides photographed. The other 47 fold-4 types remain included in the single-photo evaluation; side coverage is shown in Figure 2b.

With one relevant identity per query, MAP corresponds to mean reciprocal rank and MAP@1 to the fraction ranked first. In contrast, GAP pools query–candidate scores before calculating average precision, while GAP@1 evaluates top-ranked predictions ordered by score. Consequently, a gain in GAP@1 can coexist with lower top-1 accuracy because the measures capture different aspects of ranking and confidence.

Runs 5–7 use the ePillID reference-access setting, permitting references for test types during training while excluding validation groups and types that match their recorded appearance. Retrieval uses the full gallery of 4,902 types (Table 4). The embedding alone (MAP 0.57–0.60) falls within the range of the published plain classifiers and remains below the strongest model. Re-ranking by imprint, which the paper's model did not use, raises MAP to 0.77–0.78 and GAP@1 to 0.921, above the paper's 0.912, while MAP@1 stays lower (69.0 vs. 92.0%). The paper's ablations associate its advantage with its training recipe (contrastive and hard-mined triplet losses with softmax and ArcFace over side-specific classes, a 2,048-d embedding) alongside its backbone. Additional diagnostic comparisons did not recover the gap through crop style, the 64-d pooling step, or local versus global pooling alone.

Table 4: ePillID benchmark (fold 4, two-sided queries). Runs 5–7 permit test-type references during training but exclude validation groups. The best published model is a ResNet-152 with compact bilinear pooling and multi-head metric learning. "+ imprint" uses OCR information the published models did not use.

	MAP	GAP	MAP@1	GAP@1	top-5
Usuyama et al. [22], best	**0.958**	**0.812**	**92.0**	0.912	–
Usuyama et al. [22], plain classifiers	.547–.727	.241–.448	35.7–56.8	.331–.551	–
Ours, run 5	0.597	0.207	45.7	0.526	77.5
Ours, run 5 + imprint	0.780	0.598	69.0	**0.921**	89.6
Ours, run 7 + imprint	0.774	0.614	67.6	0.916	89.5

4.6 Error analysis and interpretation

Imprint visibility. A recorded imprint token is recovered on about 47% of validation photographs, with debossed imprints, small pills and round pills presenting the greatest difficulty. Source visibility labels also identify many unreadable views (Figure A6), indicating that the available text evidence is constrained by the images themselves.

Look-alike medicines. When two strengths of one ingredient are enrolled together, one is named as the other in 0.11% of queries, rising to 0.58% when shape and colour also match. For pairs sharing an ingredient, the imprint check reduces retakes from 7.5% to 5.2% and increases correct naming from 80.1% to 82.3%, with a small increase in wrong names. Shared manufacturer prefixes create concentrated errors: tramadol 50 mg and carvedilol 12.5 mg are confused in 5.8% of the corresponding queries despite the imprint check (Figure A7). Strength therefore remains associated with the dispensing label rather than confirmed by appearance.

Reader adaptation. Two LoRA adaptations improve validation imprint recovery from 44.0% to 53.5% and from 45.0% to 50.7%, respectively. The first produces memorised imprint strings on new pills. The second improves catalogue identification but increases counts of wrong names in the evaluated galleries from 30 to 61. Both adaptations are rejected, as are matching of visually similar characters and separate imprint scoring for each candidate. Full results and representation diagnostics are retained in Appendix D, Table A5.

5 Conclusion

PharmaScan combines a learned pill embedding, enrolment prototypes and imprint evidence to verify medicines with explicit rejection and retake decisions. The hybrid visual encoder enables enrolment of previously unseen medicines from three photographs. In the strict C3PI evaluation, the gallery using embeddings alone achieves 76.0% correct naming and 0.04% accepted misidentification among enrolled queries while rejecting 94.8% of foreign queries, and imprint re-ranking increases catalogue top-1 accuracy from 71.6% to 80.2%. Together, the experiments show that useful embedding separation depends on the training recipe and that improved imprint recovery does not necessarily reduce accepted errors.

Acquisition transfer remains the main limitation because representation learning relies on C3PI consumer images and catalogue coverage is restricted to US products. On MEDISEG, rejection of 70.4% of enrolled queries limits useful recognition despite rare misidentification among enrolled queries. Small pills and faint or debossed imprints further restrict the available evidence, with an imprint token recovered in about 47% of validation photographs. The remaining ePillID performance gap and the absence of evaluation on real phone photographs of Singapore medicines and dispensing labels also delimit the current findings.

Future work centres on local enrolment with calibration for each patient or device, evaluating accepted errors alongside rejection and retakes. Controlled branch and fusion comparisons and the ePillID multi-head recipe extend the representation analysis, while adaptation of the imprint reader is assessed through text recovery and downstream verification errors. Prospective studies extend this evaluation to usability and medication errors.

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A Data inventory and evaluation subsets

The count reconciliation in Table A1 separates repeated file-list entries, changes in identity units and deliberate exclusions from individual evaluation subsets. All 40,587 distinct C3PI consumer photographs are retained after successful cropping, and their partition counts sum to the same total. The earlier shortfall arose from the initial cropper and was resolved by re-cropping after its correction on 7 October 2026.

Table A1: Source counts and counts used in the study. Differences reflect duplicate entries, identity consolidation or evaluation eligibility. MEDISEG enrolment instances are reserved from the query set.

Unit	Source	Used	Explanation
C3PI consumer entries	40,882	40,587	295 repeated entries removed
NDC11 plus part	2,432	2,400	Package variants consolidated as NDC9 plus part
Appearance groups	2,400	2,338	111 pill types combined into 49 shared groups
Imprint-analysis photos	7,111	7,068	43 photos lack recorded imprint text
MEDISEG query pills	14,069	13,973	96 instances reserved for enrolment
ePillID two-sided types	192	145	47 types have photographs of one side only

Product identities and appearance groups. The 2,432 combinations of NDC11 and product part reduce to 2,400 pill types when package codes are consolidated. For example, Avelox 400 mg appears under two package codes although the tablet is unchanged (Figure A2a). Further consolidation by recorded imprint, shape, colour and size combines 111 pill types into 49 shared groups, yielding $2,400 - 111 + 49 = 2,338$ appearance groups. Partitioning uses these groups, so types with matching recorded appearance are kept together (Figure A1). The original distribution plot is retained in Figure A3.

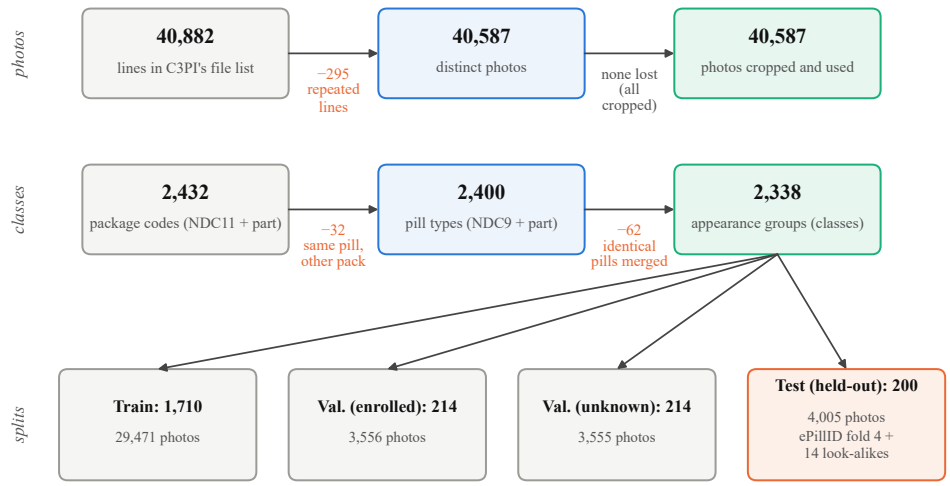


Figure A1: C3PI processing from the source file list to pill types, appearance groups and partitions. Removal of repeated entries leaves 40,587 distinct photographs, all of which are successfully cropped and assigned to partitions. Consolidation of package variants and matching appearance changes the identity count without discarding photographs.

Imprint-analysis eligibility. Of 7,111 validation photographs, 43 belong to three medicines without recorded imprint text and are excluded only from the imprint-recovery analysis. Staxyn 10 mg and a magnesium oxide 400 mg tablet carry no imprint, while Prandin 2 mg carries a logo rather than characters (Figure A2c). The resulting 7,068 photographs form the denominator of the stratified

recovery analysis in Figure A6, while all validation photographs remain available for the other evaluations.

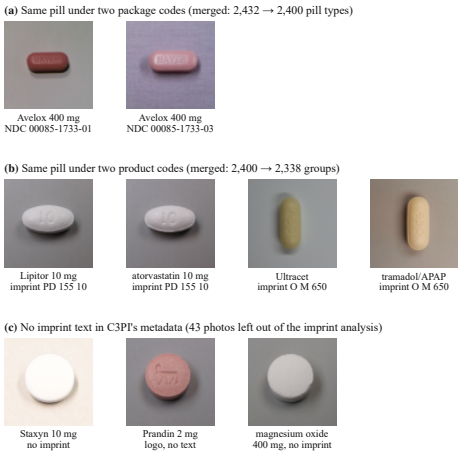


Figure A2: C3PI examples underlying identity consolidation and imprint-analysis eligibility. (a) The same tablet under different package codes. (b) Products with matching recorded appearance grouped into one class. (c) Three validation medicines without recorded imprint text, including a tablet carrying a logo.

MEDISEG enrolment and queries. The released archive contains 8,261 images and 14,069 annotated pills. Evaluation with all 32 products enrolled reserves three single-pill photographs per product, leaving $14,069 - 32 \times 3 = 13,973$ query instances. The source paper reports 8,262 images [1], one more than the released archive, whose images are all retained. The 13,973-query count concerns this evaluation with all products enrolled. Simulated personal galleries select enrolment photographs separately for each gallery, as described in Section 4.1.

ePillID query eligibility. Figure 2b summarises the availability of paired views underlying the evaluation subset in Section 4.5. Missing views restrict eligibility for the two-sided metric without removing those medicines from the single-photo evaluation.

C3PI holdout expansion. The ePillID holdout types correspond to 189 product codes and 204 C3PI pill types. Reserving another 14 types with matching recorded appearance gives 218 pill types in 200 C3PI test groups, containing 4,005 photographs.

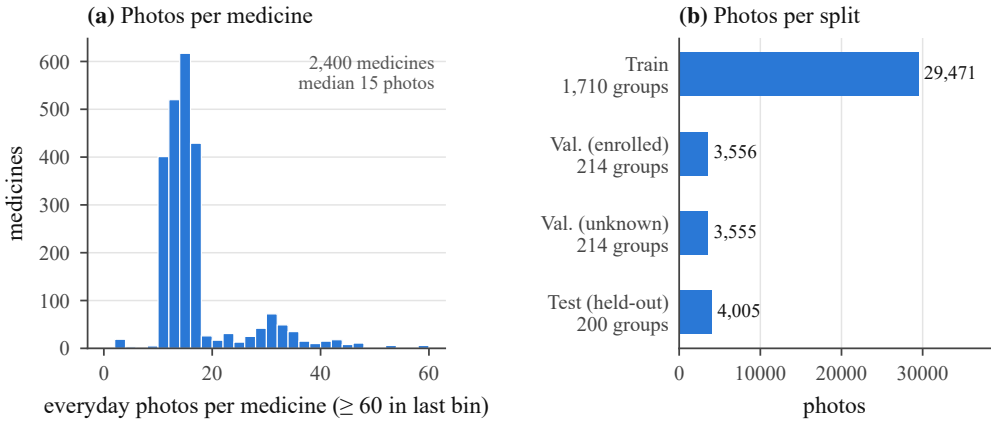


Figure A3: C3PI consumer-image distribution and partitions. (a) Images per pill type, defined by NDC9 and product part, with counts of 60 or more pooled in the final bin. (b) Partition sizes and appearance-group counts. All 40,587 distinct photographs are included, and groups containing an ePillID fold-4 holdout identity are reserved for testing.

B Architecture and training configuration

The main training schedule and augmentation are specified in Section 3.2. Table A2 lists the layer configuration during inference. Dense layers and the CBAM spatial convolution use Glorot-uniform weights with zero biases, while learned token positions are initialised uniformly in $[-0.05, 0.05]$.

Table A2: The network's layers (inference mode). Pretrained weights: VGG16 on ImageNet, ViT-B/16 on ImageNet-21k.

Block	Details	Output
CNN branch	VGG16 conv. blocks → CBAM → global average pool → Dense, ReLU	256
ViT branch	ViT-B/16 (`vit-base-patch16-224-in21k`), pooler output	768
Capsule fusion	per branch: Dense 256, ReLU, dropout 0.4, Dense 256, batch norm; dynamic routing, 2 iterations, residual update	256
Refinement	4 tokens × 64 + learned positions; 2 post-norm encoder layers (4 heads with 64 dimensions each, feed-forward width 128, dropout 0.4); mean pool	64
Embedding head	Linear, ℓ_2 normalisation	256
Auxiliary heads	shape (softmax), colour (multi-label sigmoid)	–

Input normalisation. Each RGB crop is converted to a floating-point tensor in $[0, 1]$ before branch-specific normalisation is applied inside the model. For channel c, the VGG16 input is $(x_c - \mu_c)/\sigma_c$, with $\boldsymbol{\mu} = (0.485, 0.456, 0.406)$ and $\boldsymbol{\sigma} = (0.229, 0.224, 0.225)$ in RGB order. The ViT input is $(x_c - 0.5)/0.5$, mapping the same crop to $[-1, 1]$. This processing is shared by training, evaluation and serving through the model's feature-extraction function.

Refinement attention. Each of the two refinement layers uses four attention heads with 64 dimensions per head for queries, keys and values. The implementation follows Keras attention semantics: a token of width 64 is projected separately to 256 query, key and value coordinates, which are split across the four heads. Attention scores are scaled by $1/\sqrt{64}$, and the concatenated 256-dimensional head outputs are projected back to token width 64 before the residual connection. Mean pooling across the four refined tokens therefore produces a 64-dimensional vector for the final embedding head.

C Calibration and training diagnostics

The main evaluation uses rejection, ambiguity and imprint re-ranking parameters selected on validation data. Calibration curves, sequential training changes and training trajectories are collected here to distinguish the selection procedure from the held-out results.

The project plan defines validation targets for a gallery of ten medicines enrolled from three photographs per medicine (Table A3). These are development criteria, not held-out performance claims. In particular, top-1 within the gallery and correct or incorrect accepted identities among enrolled queries are distinct measures. Held-out results are reported separately in Table 3.

Table A3: Validation targets from the project plan and reported measurements for run 4. These training-time diagnostics use validation galleries and a threshold set for 95% foreign-query rejection, without the final retake margin. Held-out results are reported separately.

Measure	Target	Run 4
Top-1 within the gallery	$\geq 95\%$	97.9%
Own-versus-foreign AUROC	≥ 0.90	0.978
Correct naming at 95% foreign rejection	$\geq 80\%$	87.9%
Misidentification rate among enrolled queries	$\leq 2\%$	1.2%

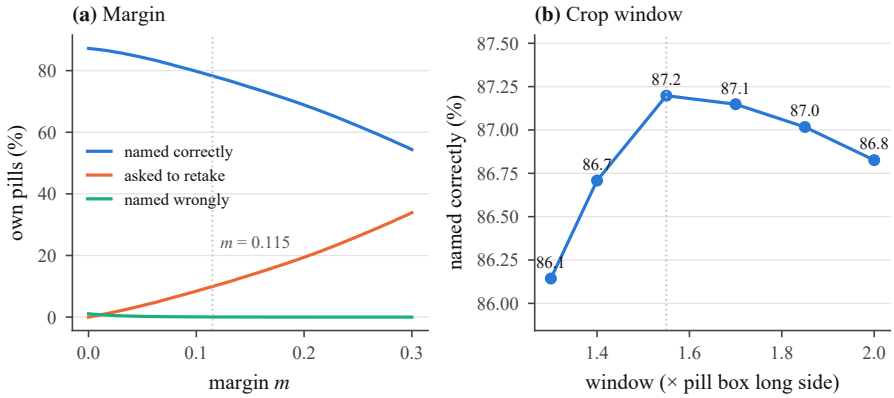


Figure A4: Settings chosen on validation (1,000 simulated patients, 10 medicines, 3 enrolment photos; τ rejects 95% of other medicines). (a) The margin trades correct names for retakes while wrong names stay near zero. (b) Correct naming across crop windows, shown with a narrow y-range.

Table A4: Sequential design comparisons and parameter sweeps. Multiple changes are combined between runs, so these comparisons do not isolate individual components. Validation results are shown unless a held-out comparison is referenced.

Change	Effect	Kept
Run 1→2: lower LRs (3×10^{-5}, 10^{-5}), separate ArcFace LR, margin warm-up	top-1 in 10 medicines 77.2→98.3%; AUROC 0.78→0.981; correct at 95% rejection 17→90.7%; dims ≈3→12–15	yes
Run 2→3: + studio reference photos	validation 98.1%; higher strict ePillID MAP@1 (Table 3)	yes
Run 3→4: + photo and augmentation with crop jitter	higher MEDISEG accuracy with lower C3PI accuracy (Table 3)	yes
Crop window 1.3–2.0×	named correctly 86.1–87.2%	1.55
Margin m 0–0.3	named correctly 87→54%; retakes 0→34%	0.115
Imprint re-ranking α, K	top-1 72.1% (α=0)→81.6%; *test* 71.6→80.2%	0.05, 100
Gallery check: distinct tokens only	named wrongly 35→30; other pills accepted 6,158→6,150	yes
Official ePillID protocol (runs 5–7)	ePillID MAP 0.57–0.60; MEDISEG 52–54%	benchmark

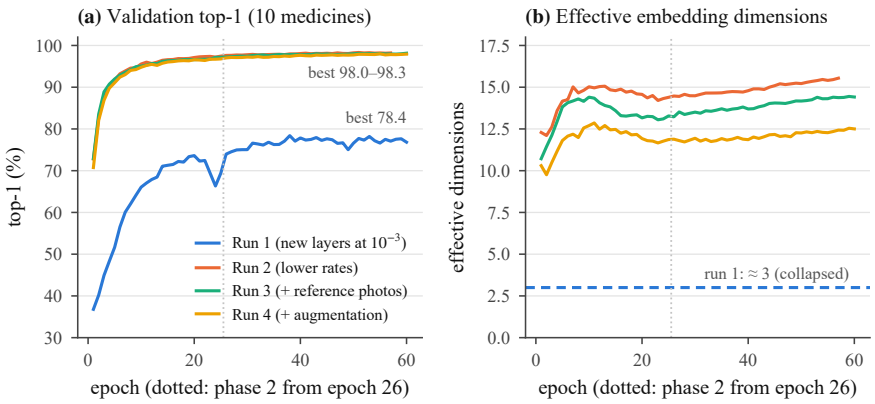


Figure A5: Training. (a) Validation top-1 within a gallery of 10 medicines never trained on. (b) Effective embedding dimensions: with the new layers at 10^{-3} the embedding collapsed to about 3.

D Additional error analysis

Stratified analysis of imprint reading. Figure A6 reports validation recovery by recorded imprint type, rater visibility, size and shape. The source analysis uses 7,068 photographs with recorded imprint text, as specified in Appendix A. The corresponding interpretation is given in Section 4.6.

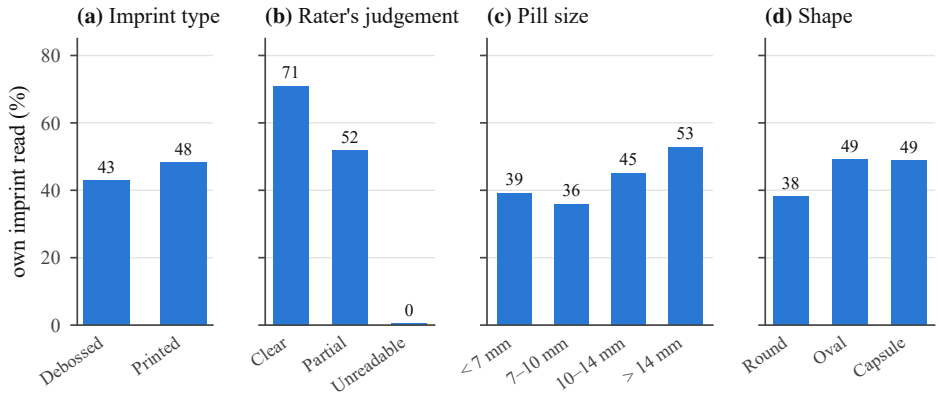


Figure A6: Share of validation photos on which a token of the pill's own imprint is read (Florence-2-large, zero-shot), by imprint type, C3PI rater's judgement of visibility, pill size and shape.

Look-alike stress tests. Both members of each pair are enrolled in the same gallery. Figure A7 compares matching by appearance alone with the imprint check for different strengths of the same ingredient and pairs from the same manufacturer. The latter share imprint prefixes, including Teva's 'TV'. Overall confusion for pairs from the same manufacturer is 0.01%, but the aggregate masks errors concentrated in individual pairs discussed in Section 4.6.

Representation diagnostics. Tracing individual queries yields capsule routing weights of approximately 9% for the CNN and 91% for the ViT. Attention between refinement tokens is nearly uniform, describing the feature use observed for these queries. The independent contributions of the branches and fusion mechanism remain unresolved because controlled architectural ablations were not performed.

Rejected variants. Table A5 retains the results of reader adaptation and other rejected variants. Its gallery entries for wrong naming are counts within their respective evaluation configurations, rather than the pooled percentages reported for held-out galleries in Table 3.

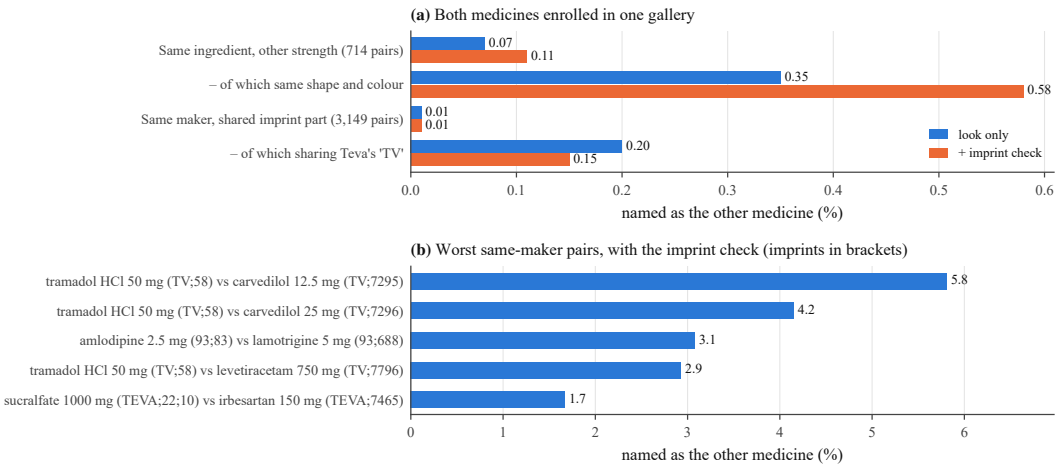


Figure A7: Look-alikes, with both medicines enrolled in one gallery. (a) Rates of wrong naming by kind of pair, look only and with the imprint check. (b) The worst pairs from the same manufacturer.

Table A5: What did not work (validation).

Tried	Result	Decision
LoRA of Florence-2-large, v1 (19,312 photos of training medicines)	imprints read 44.0→53.5%, but memorised imprints written onto new pills; identification and gallery check worse	rejected
LoRA v2 (also learns to output nothing on unreadable photos)	own imprint read 45.0→50.7%; identification 82.0→83.0%; gallery wrong names 30→61	rejected
Look-alike characters (O/0, B/8) treated as equal	measured and dropped	rejected
Scoring each candidate's imprint separately	measured and dropped	rejected
Runs 6–7: stronger fine-tuning; training aligned with the benchmark	better validation and MEDISEG, not ePillID	not served

E Implementation and project resources

The implementation environment reported in the project materials is Python 3.13.15, PyTorch 2.11.0 with CUDA 12.8, and Hugging Face Transformers 5.19.0. Training uses one NVIDIA RTX 4080 SUPER with 16 GB memory. The pipeline exposes separate stages for partitioning, training, gallery calibration, held-out evaluation, catalogue re-ranking and confidence calibration. The corresponding scripts write configuration, calibration and evaluation records to the run directory.

The model implementation uses PyTorch and Hugging Face Transformers[3], with reused project and benchmark code identified in Section 3. The research-paper-writing skill was used for language polishing.

With the project environment activated, the following PowerShell commands are run from `epills/`. They reproduce the documented sequence when the required datasets, reference catalogue and cached OCR outputs have been prepared as described in the repository.

```
$run = 'outputs/runs/stage2_run4'
python scripts/make_splits.py
python scripts/train_stage2.py --config configs/stage2_run4.yaml
python scripts/calibrate_stage2.py --run $run `
  --windows 1.3 1.4 1.55 1.7 1.85 2.0
python scripts/evaluate_stage2.py --run $run
python scripts/tune_identify_rerank.py --run $run --ocr large_c92
python scripts/evaluate_identify_topk.py --run $run --ocr large_c92
python scripts/calibrate_identify.py --run $run --ocr large_c92
```

[3] https://github.com/huggingface/transformers

```
python scripts/make_paper_figures.py
```

E.1 Project resources and application prototype

The image model is deployed through the Hugging Face interface listed in Table A6. A separate care agent application is being developed around this model service, with its deployment and source maintained independently. Its deployment demonstrates application integration in progress. The recognition results in this report evaluate the image pipeline, rather than the agent's medication management behaviour.

Table A6: Project resources supplied by the authors. The care agent application remains under development, and these deployment links document the available demonstrations.

Resource	Link
Model demo	https://davidpansx-pharmascan-api.hf.space/ui/
Model source	https://github.com/398-21/ultrasound_Model
Care agent demo	https://medithread.vercel.app
Care agent source	https://github.com/SpectateurLinlan/pharmascan-care-agent